

DAEN 429 Project Report

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Introduction

This project evaluates sign-language recognition models using static American Sign Language (ASL) alphabet classification. Four training policies were tested: T-A, T-B, T-C, and S-A. The goal is to compare model accuracy, F1 score, and generalization performance across different training strategies.

Dataset Overview

The 29-class ASL Alphabet was automatically downloaded from Kaggle. There were 87,000 training images, 28 test images, and 27 custom test images. The 29 classes consist of letters A-Z, “del”, “nothing”, and “space.”

To begin training, an 80/20 train/validation split was created, and class labels were identified. To improve generalization, we implemented data augmentations only on the training set. We implemented horizontal flips, rotations, translations, adjusted color and brightness, and added a Gaussian blur. The validation and test sets received no augmentations to ensure consistent evaluations.

Ablations Summary

Four training policies were tested in this project: T-A (head only), T-B (last block + head), T-C (progressive unfreezing), and S-A (training from scratch). The trainable parameters in each of these blocks were 14,877 in T-A, 8,408,605 in T-B, 10,508,317 in T-C, and all 11,191,389 for S-A.

Figure 1, *Figure 2*, and *Figure 3* below show the training and validation loss, accuracy, and F1 across all epochs. *Table 1* displays the highest performing epoch for each model. The results show that T-C performs the best, having the lowest loss, highest accuracy, and highest F1.

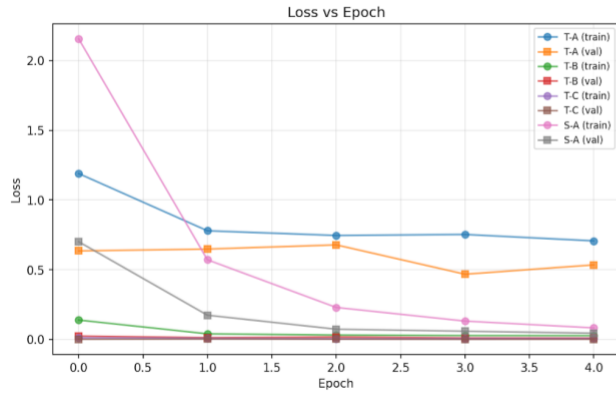


Figure 1: Loss versus epoch

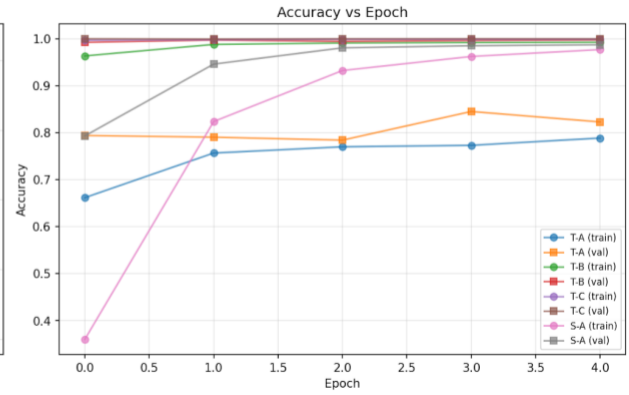


Figure 2: Accuracy versus epoch

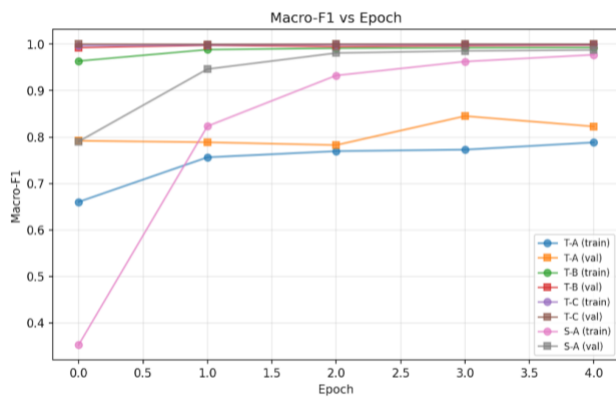


Figure 3: Macro-F1 versus epoch

Table 1: Comparison of policies and validation metrics

Policy	Val Loss	Val Acc	Val F1
T-A	0.5344	0.8228	0.8225
T-B	0.0096	0.9975	0.9975
T-C	0.0012	0.9998	0.9998
S-A	0.0444	0.9872	0.9872

Results

The first test set that was used came from the Kaggle dataset. The T-C model correctly guessed all of the images, achieving 100% accuracy. This is shown in *Figure 4*.

The custom test set was created by taking our own pictures of us doing the ASL letters. We decided to try all of the letters, as well as “nothing.” As shown in *Figure 5*, the model only achieved an accuracy of 0.5926.

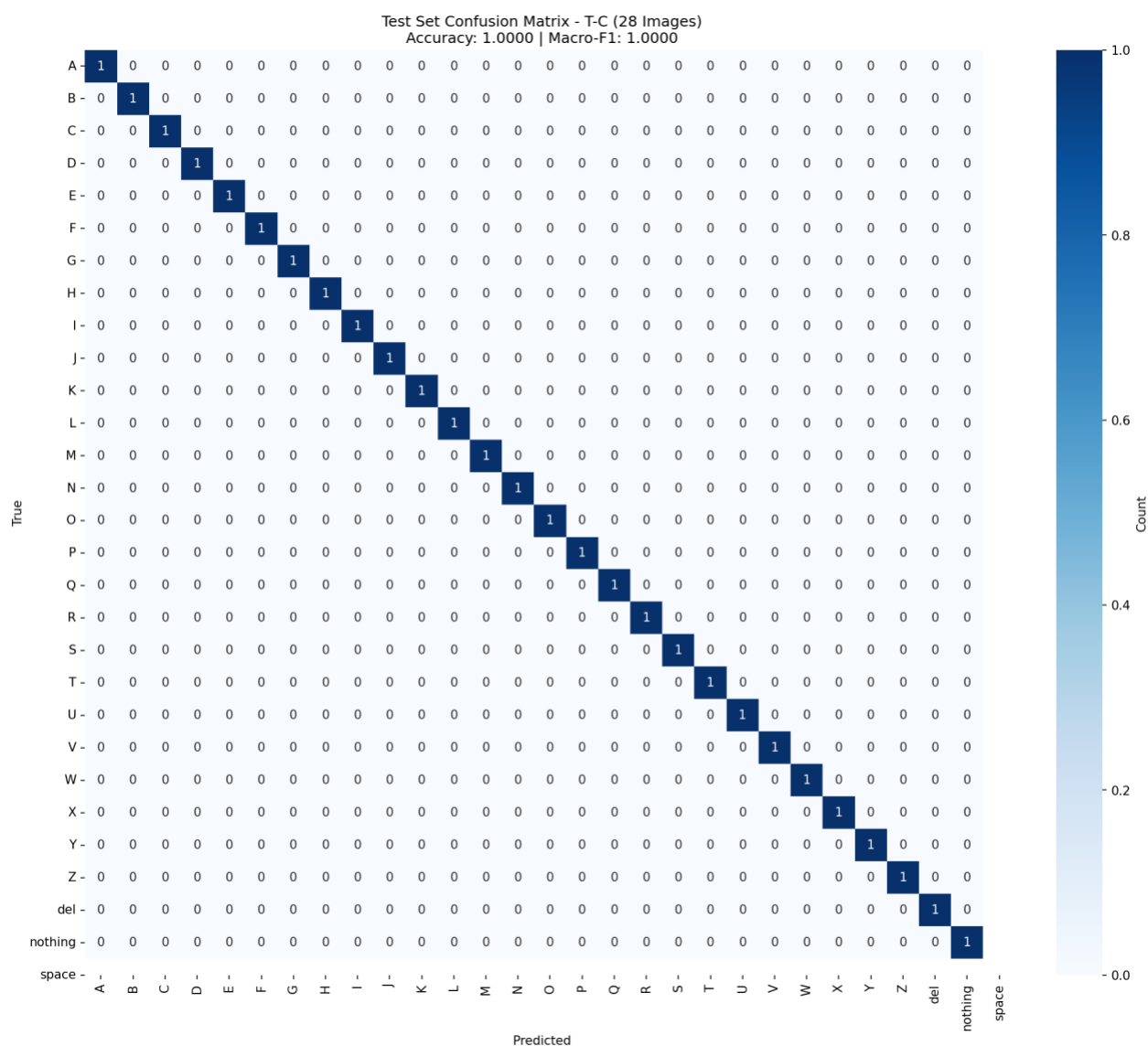


Figure 4: Test set confusion matrix

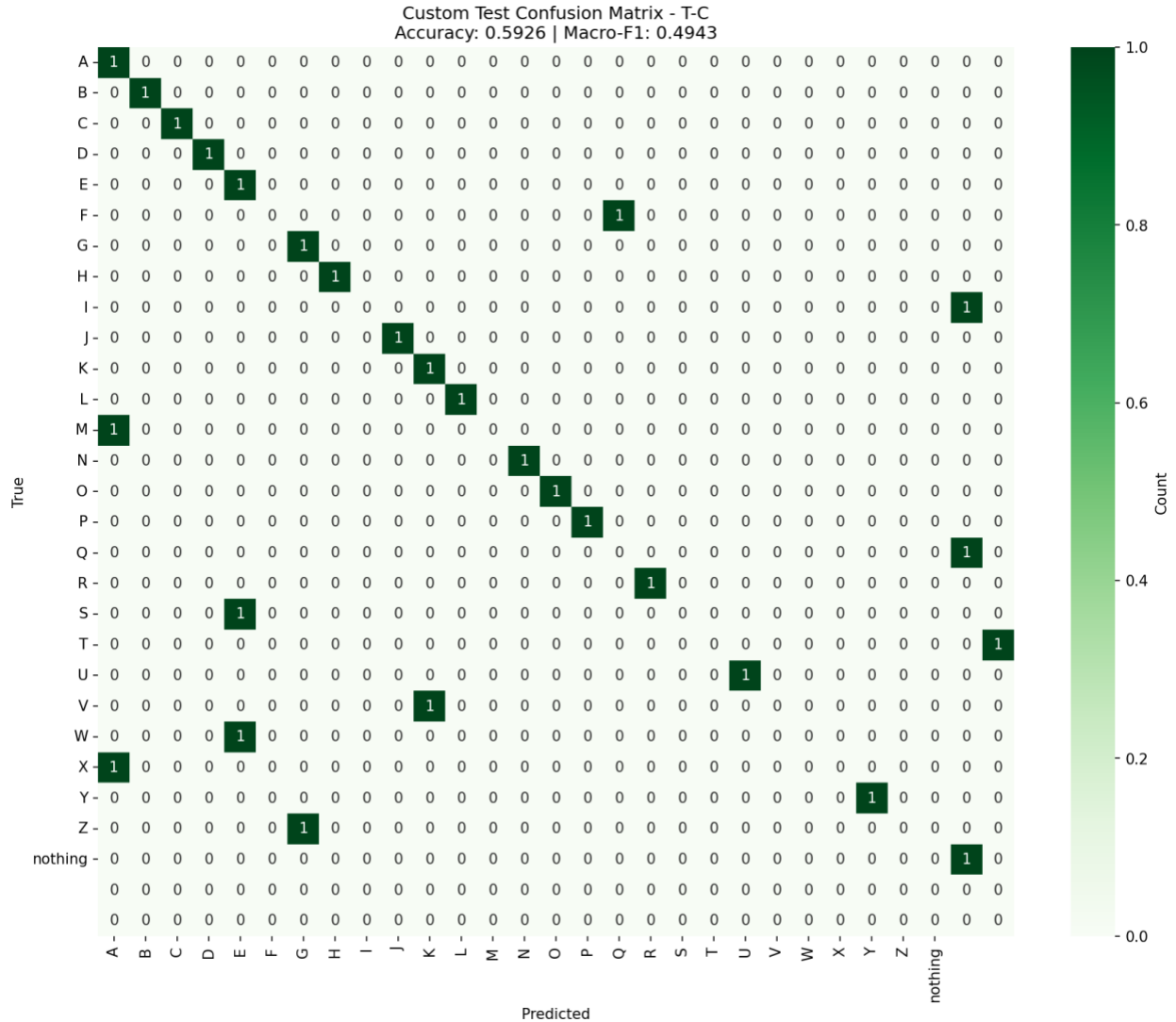


Figure 5: Custom test set confusion matrix.

Analysis

T-A achieved moderate validation accuracy (~82%), but plateaued quickly, showing limited capacity to adapt beyond the frozen feature extractor. T-B improved dramatically (~99.7% accuracy), showing that unfreezing deeper layers allows the model to learn domain-specific features. T-C achieved near-perfect validation performance (~99.98%), confirming that gradual layer unfreezing maximizes transfer learning benefits.

S-A (training from scratch) performed well (~98.7%), but required more parameters and training time, and still did not perform as well as T-C.

We determined that the best model to be used for testing was T-C. On the Kaggle test set, T-C achieved 100% accuracy. However, on the custom test set, accuracy dropped to 59.3% and F1 to about 0.49. This gap suggests that the model may be sensitive to lighting, angles, and backgrounds not present in the Kaggle dataset that were present in our dataset. Also, there did not seem to be any variance in the person who was making the signs in the images, further limiting the model's ability to generalize. For example, *Figure 6* shows a comparison of the two different test sets. There are clear differences in lighting, quality, and angle that may have caused the model to perform less accurately on the custom images.

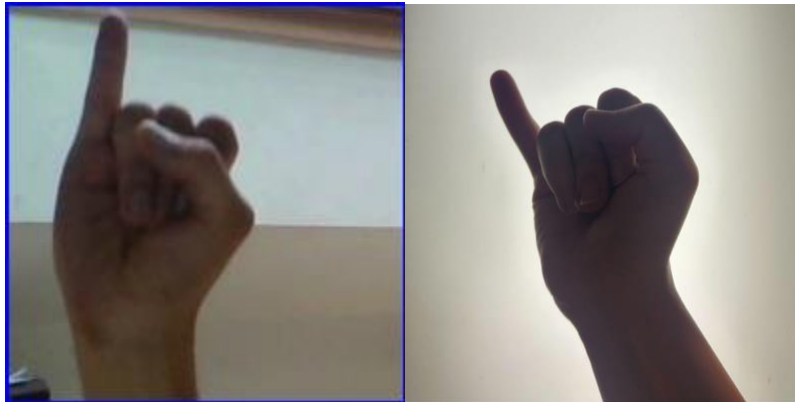


Figure 6: Letter "I" from Kaggle test set (left) and our custom test set (right).

Additionally, as we can see from the confusion matrix, most misclassifications occurred between visually similar letters, such as G vs. Z or K vs. V, highlighting the model's difficulty in distinguishing between subtle handshape variations.

Conclusions

Based on our analysis, T-C (progressive unfreezing) was the most effective model, achieving the highest validation accuracy/F1 and perfect Kaggle test set performance. Transfer learning with progressive unfreezing provided the best balance between accuracy, training stability, and computational efficiency. While training from scratch (S-A) achieved strong results, transfer learning with progressive unfreezing converged faster and reached higher accuracy, making it more efficient overall.

The drop in accuracy on the custom test set highlights the need for larger and more diverse training datasets, and possibly more adaptation techniques. We did use slight augmentation, but in the future, we may want to use stronger or more diverse augmentation strategies.

Our final takeaway is that transfer learning with careful freezing/unfreezing is highly effective for static alphabet recognition. However, real-world deployment will require addressing variability in lighting, angles, and signer differences to ensure robust performance.

Github repo: <https://github.com/sydflake/DAEN429-Project.git>